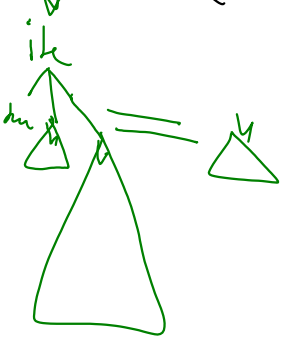
Razor

$Ty : Set$
$Tm : Ty \to Set$
$Bool : Ty$
$true, false : Tm\ Bool$
$ite : Tm\ Bool \to Tm\ A \to Tm\ A \to Tm\ A$
$ite\beta_1 : ite\ true\ u\ v = u$
$ite\beta_2 \quad false \quad v$
$Nat : Ty$
$m : \mathbb{N} \to Tm\ Nat$
$_+_ : Tm\ Nat \to Tm\ Nat$
$isZero : Tm\ Nat \to Tm\ Bool$
$+\beta : m\ n + m\ m = m\ (n+m)$
$isZero\beta_1 : isZero\ (m\ 0) = true$
$2 : \quad (m\ (1+n)) = false$

$ite\ (isZero\ (m\ 1))\ (m\ 3)\ (m\ 1 + m\ 1)$
$isZero\beta_2 = ite\ false\ (m\ 3)\ (m\ 1 + m\ 1)$
$ite\beta_2 = m\ 1 + m\ 1$
$+\beta = m\ 2$

$m\ 1 + (m\ 2 + m\ 3) = +\beta$
$m\ 1 + m\ 5 = +\beta$
$m\ 6$



$m\ 1 = m\ (1+0)$
$1 = 1+0$
$m\ 1 + m\ 2 = m\ 3$

$height : Razor.I.Tm\ A \to \mathbb{N}$
$height := M.[\![_]\!]$
where $M : Razor.Modell$
~~$height\ (m\ 1 + m\ 2) = 1$~~
~~$height\ (m\ 3) = 0$~~

$t : I.Tm\ Bool$

$t = false$ vagy $t = true$

$t : I.Tm\ Nat$

$\exists n : \mathbb{N},\ t = m\ n$

Razor nyelv normál formái

$Nf : I.Ty \to Set$
$Nf\ Bool := 2$
$Nf\ Nat := \mathbb{N}$

$\ulcorner_\urcorner : Nf\ A \to I.Tm\ A$
$\ulcorner tt \urcorner := true$
$\ulcorner ff \urcorner := false$
$\ulcorner n \urcorner := m\ n$

$Nf\ A \subseteq Tm\ A$
normalizálás:
$Nf\ A \cong I.Tm\ A$

$C : Set$
$_+_ : C \to C \to C$
$ass : (x+y)+z = x+(y+z)$
$\emptyset : C$
$idl : \emptyset + x = x$
$idr : x + \emptyset = x$
$\eta : A \to C$

monoid: $\mathbb{N}, +, 0$
$\mathbb{Z}/0, *, 1$
$(List_{Bool}) ++, []$
$((1,2,3)++(3,5))++(4,7)$
$(\quad)$
$Mátrix_{m \times n}, *, 1$

A felső sztrad monoid

norm

teljes

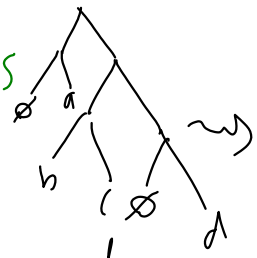
$I.C \cong \text{List } A$

stabilitás

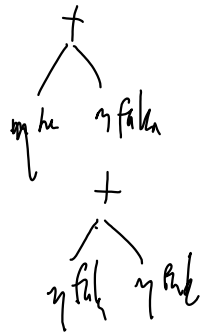
$[a, b, c, d, \ldots]$

quote, idézés

1. ∅-kat eltüntet $A = \text{Bool}$

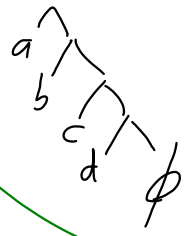


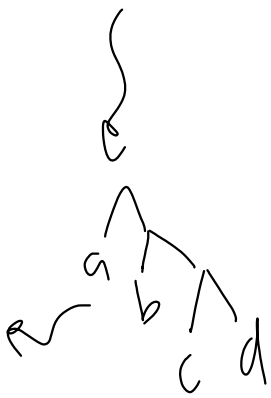
$\eta\ \text{true}$
$\eta\ \text{false}$



2. jobbra rendez

3. végére ∅-t le





norm : $I.\text{Tm } A \cong \text{Nf } A$: ⌜⌝

$\|$

$\text{St}.[\![-]\!]$

$\text{Nf Bool} = 2$
$\text{Nf Nat} = \mathbb{N}$

$? : I.\text{Tm Bool} \to I.\text{Tm Bool}$

$? : 2 \to 2$

$+\beta$
$\text{istrue}\beta_{1,2}$
$\text{iszero}\beta_{1,2}$

Stabilitás : $\{A\}(n : \text{Nf } A) \to \text{norm}\ \ulcorner n \urcorner = n$

stab $\{\text{Bool}\}$ tt : $\text{norm}\ \ulcorner tt \urcorner \overset{?}{=} tt$ $\overset{\text{def}}{=}$

$\|\text{def}$

stab $\{\text{Bool}\}$ ff

stab $\{\text{Nat}\}$ n : $\text{norm}\ \ulcorner n \urcorner = \text{norm}(\text{num } n) = \text{St.num } n \overset{\text{def}}{=} n$

$\text{norm true} \overset{\text{def}}{=} \text{St}.[\![\text{true}]\!] \overset{\text{def}}{=} \text{St.true}$

Compl : $(t : I.\text{Tm } A) \to \ulcorner \text{St}.[\![t]\!] \urcorner = t$

D : DepModel

$\text{Ty}^\bullet\ A := T$

$\text{Tm}^\bullet\ \{A\}\ _\ t := (\ulcorner \text{St}.[\![t]\!] \urcorner = t)$

$\text{true}^\bullet : \boxed{\text{Tm}^\bullet\ \{\text{Bool}\}\ _\ \text{true}}$

$= \ulcorner \text{St}.[\![\text{true}]\!] \urcorner \overset{?}{=} \text{true}$

$\|\text{def}$

$\ulcorner tt \urcorner$

def

$\text{istero}^\bullet\ t^\bullet : \ulcorner \text{St}.[\![\text{istero } t]\!] \urcorner \overset{?}{=} \text{istero } t$ $t^\bullet$

$\|\text{def}$

$\ulcorner \text{St.istero}\ [\![t]\!] \urcorner = \text{istero}\ \ulcorner [\![t]\!] \urcorner$

$\text{istero}\ \ulcorner\ \urcorner$

$\ulcorner \text{St.istero } n \urcorner = \text{istero}\ \ulcorner n \urcorner$

$n = 0 \to \ulcorner \text{St.istero } 0 \urcorner \overset{\text{def}}{=} \ulcorner tt \urcorner \overset{\text{def}}{=} \text{true} \overset{\beta 1}{=} \text{istero}(\text{num } 0) \overset{\text{def}}{=} \text{istero}\ \ulcorner 0 \urcorner$

$n = 1 + m \to \ulcorner \text{St.istero}(1+m) \urcorner \overset{\text{def}}{=} \ulcorner ff \urcorner \overset{\text{def}}{=} \text{false} \overset{\beta 2}{=} \text{istero}(\text{num}(1+m)) \overset{\text{def}}{=} \text{istero}\ \ulcorner 1+m \urcorner$

$\underbrace{u^\circ}_{:\ \ulcorner[\![u]\!]\urcorner = u} +^\circ \underbrace{v^\circ}_{\ulcorner[\![v]\!]\urcorner = v} : \ulcorner[\![u +^\circ v]\!]\urcorner \overset{?}{=} u +^\circ v$

$\| \qquad u^\circ, v^\circ$

$\ulcorner[\![u]\!] +^m [\![v]\!]\urcorner = \ulcorner[\![u]\!]\urcorner + \ulcorner[\![v]\!]\urcorner$

$\uparrow$

$\ulcorner n + m \urcorner \overset{?}{=} \ulcorner n \urcorner + \ulcorner m \urcorner$

$\| \text{def} \qquad \| \text{def}$

$m(n+n) \underset{+\beta}{=} m\,n + m\,m$

Razor $\mathbb{N}, \mathbb{Z}$

$t = \text{ite}(\text{isZero}(n\,1 + n\,2))$

$u = \text{true}$

$t \overset{?}{=} u$

$\text{canon} \| \qquad \| \text{def}$

$\ulcorner \text{nom}\, t \urcorner = \ulcorner t \urcorner$

False

köv. óra 9 percel korábbi